THE RELOCALIZATION SQUARE OF RINGED TOPOI: A MORPHISM OF RINGED TOPOI RELOCALIZES ALONG A MORPHISM OF SHEAVES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file sites-modules.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites two external items (the ringed-topos relocalization lemma and a section), recorded in assembly.external_references with canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- Localization of a morphism of topoi along a morphism of sheaves. Stacks Project, Sites, Lemma sites-lemma-localize-morphism-topoi (tag 00YW). Supplies the diagram of underlying topoi, the identity f'_* j_F^{-1} = j_G^{-1} f_* and the description of (f')^{-1}.
- Definition of a morphism of ringed topoi. Stacks Project, Modules on Sites, section on morphisms of ringed topoi (structure map of a morphism). Gives the data that must be verified to conclude that the relocalized square is a square of ringed topoi.

Source reference. Original reference 04H1: sources/stacks-chow/tags.txt, source lines 5852–5852. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 04IE: sources/stacks-chow/tags.txt, source lines 5900–5900. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let*

$$f : (Sh(\mathcal{C}), \mathcal{O}) \longrightarrow (Sh(\mathcal{D}), \mathcal{O}')$$

be a morphism of ringed topoi. Let $\mathcal{G}$ be a sheaf on $\mathcal{D}$. Set $\mathcal{F} = f^{-1}\mathcal{G}$. Then there exists a commutative diagram of ringed topoi

$$\begin{array}{ccc} (Sh(\mathcal{C})/\mathcal{F}, \mathcal{O}_\mathcal{F}) & \xrightarrow[(j_\mathcal{F}, j_\mathcal{F}^\sharp)]{} & (Sh(\mathcal{C}), \mathcal{O}) \\ \downarrow{\scriptstyle (f', (f')^\sharp)} & & \downarrow{\scriptstyle (f, f^\sharp)} \\ (Sh(\mathcal{D})/\mathcal{G}, \mathcal{O}'_\mathcal{G}) & \xrightarrow{(j_\mathcal{G}, j_\mathcal{G}^\sharp)} & (Sh(\mathcal{D}), \mathcal{O}') \end{array}$$

We have $f'_* j_\mathcal{F}^{-1} = j_\mathcal{G}^{-1} f_*$ and $f'_* j_\mathcal{F}^* = j_\mathcal{G}^* f_*$. Moreover, the morphism f' is characterized by the rule

$$(f')^{-1}(\mathcal{H} \xrightarrow{\varphi} \mathcal{G}) = (f^{-1}\mathcal{H} \xrightarrow{f^{-1}\varphi} \mathcal{F}).$$

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Proof. By Sites, Lemma 04H1 we have the diagram of underlying topoi, the equality $f'_* j_{\mathcal{F}}^{-1} = j_{\mathcal{G}}^{-1} f_*$, and the description of $(f')^{-1}$. To define $(f')^\sharp$ we use the map

$$(f')^\sharp : \mathcal{O}'_{\mathcal{G}} = j_{\mathcal{G}}^{-1}\mathcal{O}' \xrightarrow{j_{\mathcal{G}}^{-1} f^\sharp} j_{\mathcal{G}}^{-1} f_* \mathcal{O} = f'_* j_{\mathcal{F}}^{-1}\mathcal{O} = f'_* \mathcal{O}_{\mathcal{F}}$$

or equivalently the map

$$(f')^\sharp : (f')^{-1}\mathcal{O}'_{\mathcal{G}} = (f')^{-1} j_{\mathcal{G}}^{-1}\mathcal{O}' = j_{\mathcal{F}}^{-1} f^{-1}\mathcal{O}' \xrightarrow{j_{\mathcal{F}}^{-1} f^\sharp} j_{\mathcal{F}}^{-1}\mathcal{O} = \mathcal{O}_{\mathcal{F}}.$$

We omit the verification that these two maps are indeed adjoint to each other. The second construction of $(f')^\sharp$ shows that the diagram commutes in the 2-category of ringed topoi (as the maps $j_{\mathcal{F}}^\sharp$ and $j_{\mathcal{G}}^\sharp$ are identities). Finally, the equality $f'_* j_{\mathcal{F}}^* = j_{\mathcal{G}}^* f_*$ follows from the equality $f'_* j_{\mathcal{F}}^{-1} = j_{\mathcal{G}}^{-1} f_*$ and the fact that pullbacks of sheaves of modules and sheaves of sets agree, see Lemma 04IE. □